

SELF-SENSING CFRP LAMINATES WITH ELECTROSPUN BARIUM TITANATE NANOFIBROUS INTERLEAVES FOR INTEGRATED STRUCTURAL HEALTH MONITORING

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Abstract

Structural Health Monitoring (SHM) of aerospace carbon-fiber laminates requires integrated sensing solutions capable of detecting barely visible impact damage while preserving structural integrity. Conventional surface-mounted piezoelectric sensors are prone to environmental degradation and debonding, whereas embedded bulk or patch sensors may act as stress raisers, reduce interlaminar strength, and locally alter laminate stiffness. Piezoelectric polymers offer good flexibility but exhibit limited electromechanical performance and reduced operating temperature due to their relatively low Curie temperature. In contrast, piezoceramics provide high piezoelectric coefficients and a wide usable temperature range at the expense of brittleness and poor mechanical compatibility with the surrounding composite.

In recent years, research has focused on piezoelectrically active composites, in which the sensing phase is distributed in the matrix, as filler, or introduced as interleaved veils. In this work, a fully integrated piezoelectric sensor is realized in a CFRP prepreg laminate by interleaving electrospun barium titanate (BaTiO₃) nanofibrous veils at the mid-plane. Additional electrospun nylon veils are employed as electrical insulators between CFRP plies, which function directly as the sensor electrodes and eliminate the need for separate metallic foils. The influence of autoclave cure pressure on BaTiO₃ nanofiber density and on the resulting piezoelectric response of the laminate is systematically assessed.

The piezoelectric laminates are characterized through polarization-electric field (P-E) loop measurements to elucidate their ferroelectric behaviour as a function of electric field amplitude, excitation frequency, and temperature. Cyclic through-thickness compression tests are then performed to determine the voltage output, sensitivity, and effective piezoelectric coefficient of the embedded sensor. Overall, the study provides an overview of the processing-structure-property relationships governing this class of BaTiO₃ nanofiber-CFRP laminates and highlights the potential of nanofibrous piezoceramic interleaves for reliable SHM of aerospace composite structures

Keywords: SHM; Electrospinning; Barium Titanate; Nylon nanofibers; CFRP

1 Introduction

The aerospace sector heavily relies on advanced carbon fiber reinforced polymers (CFRPs) to achieve lightweight and fuel-efficient structures. However, CFRPs are highly susceptible to Barely Visible Impact Damage (BVID), such as internal delamination, which can drastically reduce the structural integrity [1]. To address this, Structural Health Monitoring (SHM) systems are implemented. Conventional SHM techniques often involve the integration of bulky sensors, such as Lead Zirconate Titanate (PZT) ceramics or optical fibers. These inclusions act as defects, creating stress concentrations that weaken the composite structure they are meant to monitor [2].

Recent advancements have shifted towards self-sensing materials, utilizing functional nanomaterials to provide sensory capabilities without degrading mechanical properties. Electrospun polymeric nanofibrous mats have emerged as excellent candidates for interleaving in CFRPs. Specifically, Nylon (PA6,6) electrospun veils have been widely proven to significantly enhance the interlaminar fracture toughness of composites, suppressing delamination growth [3].

In this work, lead-free Barium Titanate (BaTiO_3 , BT) nanofibers were electrospun and calcined to serve as the active sensing phase. To prevent short circuits between the conductive CFRP plies (acting as electrodes), a Nylon-6,6 electrospun veil was added as a dielectric spacer. This study examines the macroscopic ferroelectric response of these novel multi-layered self-sensing laminates, analyzing P-E hysteresis loops under various thermal and electrical conditions. Furthermore, this study establishes a correlation between manufacturing conditions (compaction pressure), the resulting interlaminar morphology (fiber density and thickness), and the macroscopic dielectric/piezoelectric response. This integrated processing-structure-property analysis provides a comprehensive framework for designing minimally intrusive SHM laminates with optimized sensing performance. The target is to achieve sufficient amount of BT nanofibers to raise the effective piezoelectric coefficient while managing the capacitive value through architectural and boundary-condition control at the laminate scale.

2 Materials and Methods

2.1 Synthesis of BaTiO_3 nanofibers

BT nanofibers were produced through a sol-gel electrospinning technique, following methodologies established in previous literature [4–6]. For the precursor preparation, 4.78 g of barium acetate ($\text{Ba}(\text{CH}_3\text{COO})_2$, Merck, Germany) was dissolved in 15.74 g of acetic acid (CH_3COOH , $\geq 99\%$, Sigma-Aldrich, USA) at 70°C for 1 hour under continuous agitation. The solution was then cooled to 5°C , after which 5.31 g of titanium (IV) isopropoxide ($\text{C}_{12}\text{H}_{28}\text{O}_4\text{Ti}$, 97%, ABCR GmbH & Co. KG, Germany) was introduced dropwise. The resulting mixture was stirred at room temperature for 15 hours to ensure a complete chemical reaction. To optimize spinnability and viscosity, the precursor was blended with a 7.8 wt% solution of polyvinylpyrrolidone (PVP, $M_w \approx 1,300,000$, Sigma-Aldrich, USA) in ethanol (Merck, Germany) in a 1:1 volume ratio, followed by an additional hour of stirring. Electrospinning was carried out using a NEU-Pro system (NaBond Technologies Co., Limited, China) equipped with a 0.218 mm metallic needle. The process parameters were maintained at a voltage of 15 kV, a flow rate of 0.3 mL/h, and a tip-to-collector distance of 10 cm, utilizing a drum collector rotating at 10 rpm. Post-processing involved calcination of the nanofiber mats at 1100°C for 1 hour (heating rate of 120°C/h) to achieve the final crystallized ceramic structure. The complete synthesis workflow is illustrated in Figure 1.

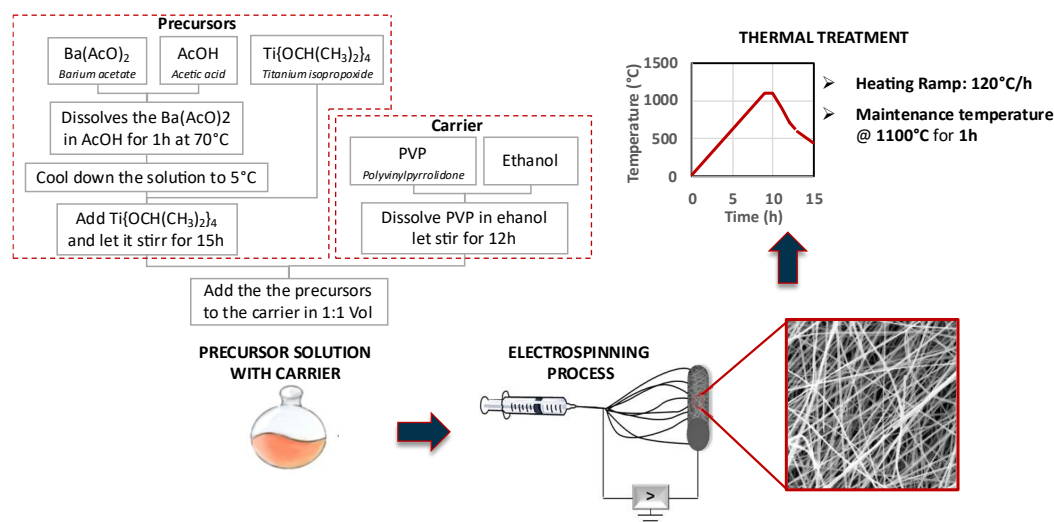


Figure 1 Workflow of the electrospinning–calcination route used to synthesize BaTiO_3 nanofibers, from precursor solution preparation to thermal treatment and final mat collection.

2.2 Fabrication of the Nylon-6,6 insulating veil

To provide the necessary electrical insulation between the active BT sensing layer and the conductive CFRP electrodes, a Nylon-6,6 nanofibrous mat was fabricated. The production was carried out using a multi-needle drum-based electrospinning system (Spinbow S.r.L.) with the following operational parameters: an applied voltage of 27 kV, a tip-to-collector distance of 15 cm, and a feed rate of 0.50 mL/h per nozzle. The process was conducted under controlled environmental conditions (24°C and 40% relative humidity) with a drum collector operating at a linear speed of 0.4 m/s. This setup resulted in nanofibrous mats with a consistent areal density of approximately 8 g/m^2 .

2.3 Laminate manufacturing

The ferroelectric behavior of the piezoactive interlayer was investigated using a laminate architecture with a final stacking sequence of $[(\text{CFRP-}90^\circ)/(\text{CFRP-}0^\circ)/\text{BT \& Nylon-mask}]_s$. To prevent electrical contact between adjacent carbon fibers acting as signal electrodes, a nanofibrous Nylon mask was placed surrounding the BT nanofiber mat at the mid-plane. To evaluate the impregnation behavior and the possibility of tailoring fiber density, four different autoclave curing cycles were performed with varying external pressures: 0, 1, 3, and 6 bar. All cycles followed the same thermal profile: an initial stage at 50°C for 30 min, followed by a second stage at 100°C for 4 h to complete epoxy cross-linking, with a heating ramp of 60°C/h . Non-sensing reference laminates were manufactured with the same architecture but replaced the BT interlayer with only the Nylon nanofibrous mat. The architectural concept and the dimensions of the internal components are exemplified in Figure 2.

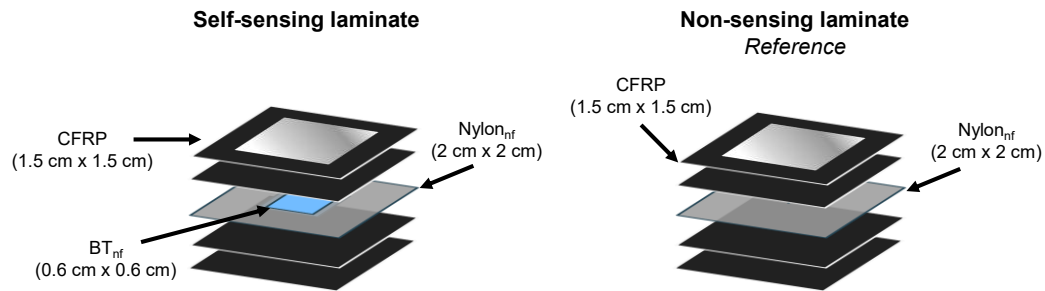


Figure 2 Schematic of the self-sensing and non-sensing (reference) laminates, with plan-view dimensions of the CFRP electrodes and of the Nylon and BT nanofiber interlayers.

2.4 Experimental Characterization

2.4.1 Polarization Analysis and P-E Loops

The ferroelectric response of the calcined fiber mats embedded into the laminate was evaluated using a TF Analyzer 2000 system (aixACCT Systems GmbH, Germany) equipped with a high-temperature piezoelectric sample holder. To establish electrical contact, laminate surfaces were gently abraded and coated with silver paste (1 cm²). The experimental campaign focused on the investigation of the surface polarization (expressed in C/m²) by applying a triangular half-wave signal. Three key parameters were varied: the applied electric field (1, 2.5, and 5 kV/mm), the specimen temperature (24, 50, 75, and 100 °C), and the P-E loop frequency (0.01 and 0.1 Hz). The remnant polarization at zero voltage (P_r) and the polarization at maximum voltage (P_{max}) were identified as primary indicators. Additionally, cross-sectional SEM micrographs were analyzed using image processing techniques to quantify the volume fraction and fiber density of the BT nanomat embedded in the composite laminate.

2.4.2 Poling Process

To induce macroscopic piezoelectricity, the embedded nanofibrous interlayer underwent a poling treatment. The process was conducted using a Keithley 6517B electrometer in voltage source mode, applying an electric field of 5 kV/mm for 10 minutes at a constant temperature of 100 °C. Following this stage, the specimens were cooled down to room temperature while maintaining the applied field to stabilize the piezoelectric state, according to the procedure defined in [7].

2.4.3 Piezoelectric Response and Sensitivity

The macroscopic piezoelectric performance was characterized using a universal testing machine (Instron 8800). The self-sensing laminates were subjected to cyclic compressive loads between -500 and -2000 N at a frequency of 20 Hz. Force and output voltage signals were simultaneously acquired at a sampling rate of 833 Hz. Sensitivity was determined as the ratio between the peak-to-peak output voltage and the peak-to-peak input force. Electrical properties (capacitance and resistance) were measured at 1 kHz using an RLC meter (GW-Instek 6002). Finally, the effective piezoelectric coefficient d_{33}^{eff} of the interlayer was extrapolated from the measured capacitance and sensitivity values, following the analytical procedure described in [7].

3 Results and Discussion

3.1 Morphological Characterization

A major advantage of the nanofibrous architecture is its intrinsic porosity, which promotes efficient resin infiltration during autoclave curing. This feature ensures seamless integration of the interlayer into the CFRP laminate, avoiding the formation of voids or compromised consolidation. This porous morphology was confirmed by microstructural analysis of both calcined BT and Nylon-6,6 nanofibers, Figure 3. Quantitative diameter analysis, highlights the significant impact of the thermal treatment. The calcined BT nanofibers reached an average diameter of 240 ± 61.3 nm, representing approximately a 50% reduction compared to the as-spun sol-gel/PVP precursors (503 ± 85.1 nm). Nylon nanofibers exhibited an average diameter of 453 ± 73.4 nm. Material choice and thermal processing thus define the final fiber morphology, directly impacting resin permeation and interlayer functionality.

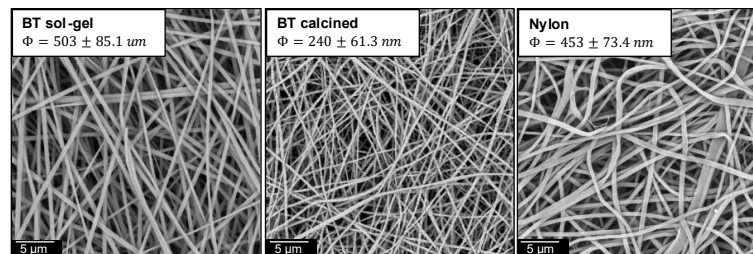


Figure 3 SEM images of the nanofibers as spun, with their corresponding average diameters and standard deviations

The cross-sectional SEM analysis of the integrated laminates confirms that the nanofibrous mats were successfully embedded without evidence of voids, achieving complete resin permeation even at 0 bar of external curing pressure. However, the curing pressure significantly influenced the internal morphology: the BT fiber volume fraction increased from approximately 17% in specimens processed at 0 bar to 30% in those cured at 6 bar. This trend results from the enhanced resin "squeezing-out" effect at higher pressures, which reduces the resin content within the interlayer and promotes fiber compaction. Consequently, an inverse correlation between curing pressure and interlayer thickness was observed, as illustrated in Figure 4 alongside the corresponding increase in fiber density.

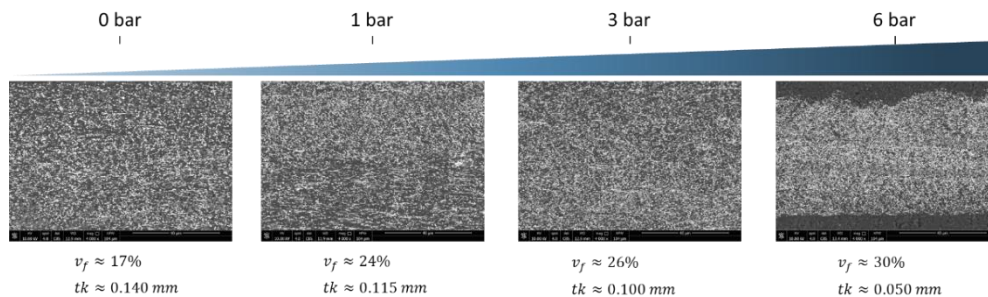


Figure 4 Cross-sectional SEM images of laminates containing BT nanofiber interlayers, showing void-free impregnation and complete resin permeation across curing pressures.

3.2 P-E loops

To clarify the underlying mechanisms, the P-E hysteresis loops of the self-sensing laminate featuring the highest fiber density (cured at 6 bar) were compared with those of the non-sensing reference laminate (Nylon-only). As shown in Figure 5, none of the samples displayed the characteristic saturated hysteresis of bulk ferroelectric materials; instead, the loops remained elliptical, indicating that the response is primarily governed by dielectric displacement. The shape of the loops was strongly influenced by temperature and frequency: at room temperature and 50 °C, the response was relatively stable, whereas at 75 °C and 100 °C, electrical losses became dominant. This led to a significant widening of the loop area, where the remnant polarization (P_r) consistently exceeded the polarization at maximum voltage (P_{max}), especially at the lower frequency of 10 mHz. A comparative analysis confirms that both self-sensing and non-sensing laminates follow nearly identical trends, supporting the hypothesis that the measured polarization originates from electric displacement and conduction currents within the polymeric components (epoxy and Nylon) rather than from intrinsic ferroelectric behavior of the BaTiO₃ phase. Specifically, above 75 °C, conduction effects become the dominant mechanism, confirming the lossy dielectric nature of the sensing architecture.

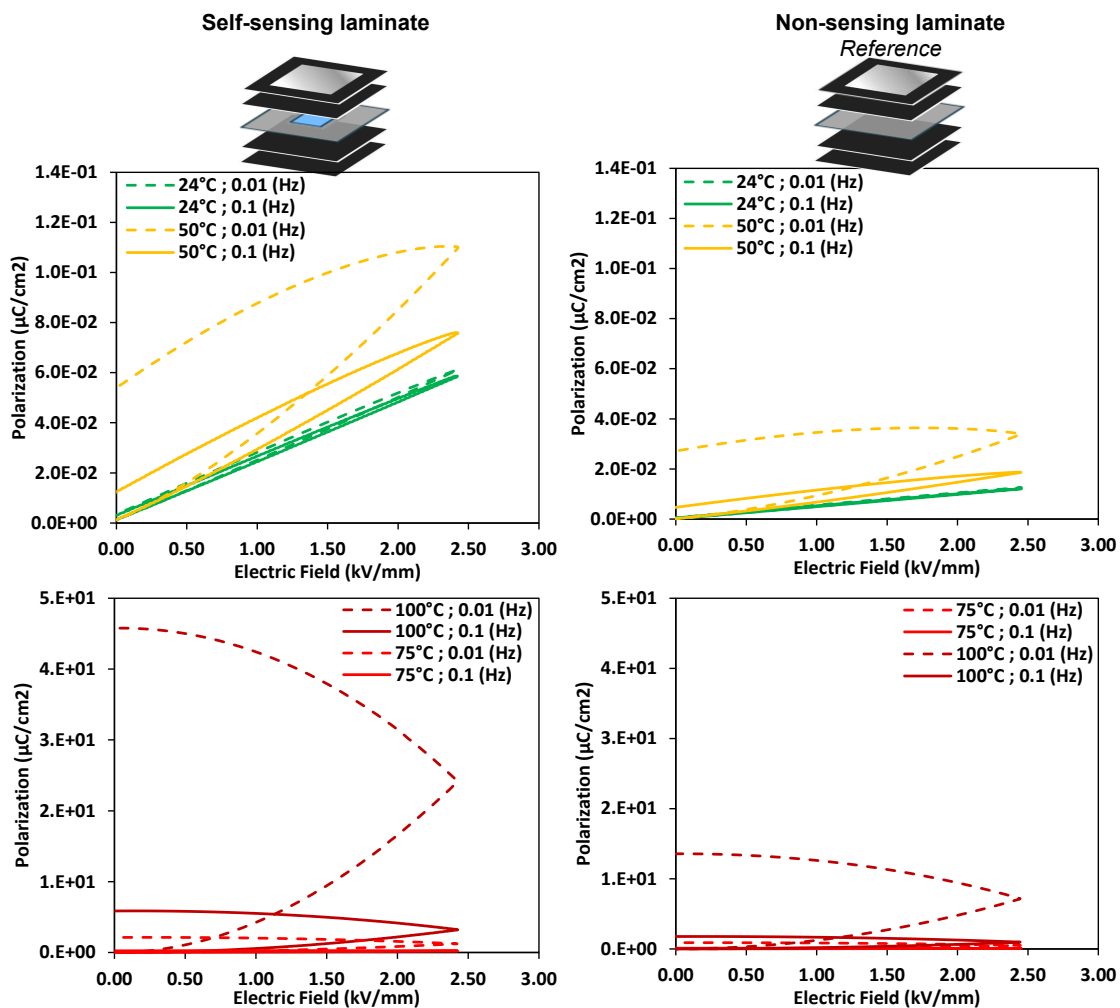


Figure 5 P-E loops for self-sensing (6 bar) and non-sensing reference laminates at different temperatures and frequencies.

3.3 Piezoelectric performances

The ability of the self-sensing architecture to generate a piezoelectric signal was evaluated by testing poled specimens under cyclic compressive loading. The output voltage and applied load signals were perfectly in phase for all tested fiber densities (17%, 24%, and 30%), confirming the piezoelectric nature of the electrical response. As shown in Figure 6, an increase in BT fiber density led to higher peak-to-peak output voltages. Although the electrical capacitance also increased with density due to reduced interlayer thickness (consistent with parallel-plate capacitor behavior), the overall electromechanical sensitivity improved. Specifically, the effective piezoelectric coefficient (d_{33}^{eff}) showed a clear increasing trend with fiber density. These results demonstrate that the laminate's performance is governed by the volume fraction and connectivity of the active piezoceramic phase within the composite architecture.

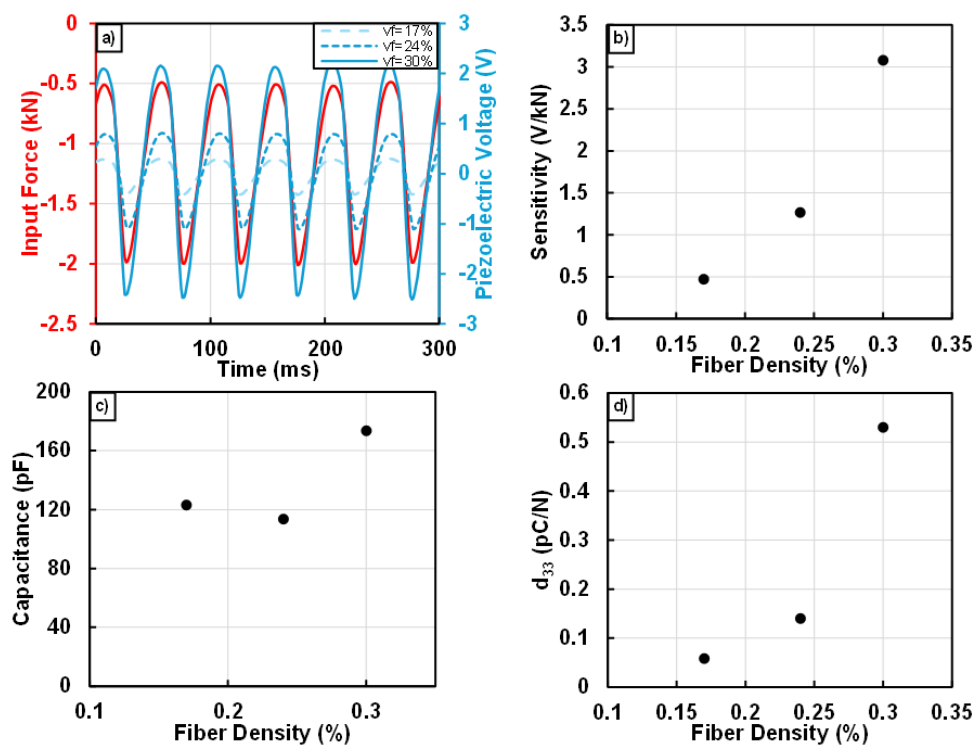


Figure 6 Piezoelectric characterization: (a) output voltage/load signals, (b) voltage vs. density, (c) capacitance, and (d) effective d_{33} coefficient.

4 Conclusions

This study demonstrates that electrospun BaTiO₃ (BT) nanofibrous interlayers can be successfully integrated into CFRP laminates, with their morphology and fiber density effectively controlled by autoclave curing pressure. Increasing the external pressure allows for the compaction of the active phase, raising the BT volume fraction from approximately 17% to 30% and enabling a void-free structure even at low pressures. While the polarization behavior is dominated by dielectric displacement with significant temperature and frequency dependent losses (particularly above 75 °C), the embedded interlayers exhibit genuine piezoelectric conversion. The macroscopic in-phase voltage output under cyclic loading confirms the functional nature of the self-sensing architecture. Crucially, higher fiber densities led to an

increase in the effective piezoelectric coefficient (d_{33}^{eff}), which more than compensated for the increase in inter-electrode capacitance, resulting in enhanced electromechanical signal amplitude. These results establish a processing-structure-property framework for SHM laminates, where performance is optimized by balancing active-phase connectivity with the electrical boundary conditions established at the laminate scale. The integrated Nylon-6,6 masks successfully prevented short-circuiting, providing a promising and non-invasive pathway for advanced, self-sensing composite structures.

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